University of Kelaniya

Faculty of Science – Department of Industrial Management

INTE 22253 - Assignment

Assessment Type: Group Assignment + Individual Report + Physical Viva

Total Marks: 100

Submission Format: PDF report (+ code/scripts where applicable, zipped separately)

Word Limit (written parts): 2,500–3,500 words (excluding code, diagrams, references)

Group Composition Rules

- Minimum students per group: 2
- Maximum students per group: 4
- Same case study topic for the entire group
- Each student MUST submit an individual report
- Viva and presentation are conducted as a group, but assessment is both group-based and individual

⚠️ Groups that violate size limits will not be evaluated.

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1 Assignment Objectives

On successful completion of this assignment, students will be able to:

- Explain the challenges of distributed transactions and coordination in multi-agent systems.
- Describe the architecture and operating principles of distributed search engines and P2P systems.
- Compare cloud programming models and evaluate their suitability for different workloads.
- Differentiate between object, block, and file storage, and explain distributed file system concepts.
- Critically analyze real-world systems (GFS, HDFS, Amazon S3) as case studies.
- Evaluate design trade-offs in building distributed file systems.

Assignment Structure

This assignment has three parts: Part A (Conceptual Questions), Part B (Case Study Analysis), and Part C (Design/Practical Task). **All parts are compulsory.**

Part A — Conceptual Questions (20 marks)

Answer all of the following questions. Each answer should be concise, well-structured, and supported with diagrams or examples where appropriate.

1. Distributed Transactions (5 marks)

   - Explain the concept of the Two-Phase Commit (2PC) protocol. Discuss its main weaknesses and how Three-Phase Commit (3PC) attempts to address them.
   - Describe how consensus protocols (e.g., Paxos or Raft) support coordination in multi-agent distributed systems. Why is consensus considered a harder problem than simple transaction commitment?
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2. Multi-Agent Systems (5marks)

   Discuss the key characteristics of multi-agent systems in a distributed environment. How do agents coordinate, communicate, and resolve conflicts when accessing shared resources?

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3. Dev. Distributed Search Engines (5 marks)
   - Explain how a distributed search engine indexes and retrieves data across multiple nodes (e.g., using inverted indexes and sharding).
   - Compare centralized search architectures with distributed/P2P search architectures in terms of scalability, fault tolerance, and query latency.

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4. P2P Systems (5 marks)

Compare structured P2P systems (e.g., Chord, Kademlia using DHTs) with unstructured P2P systems (e.g., Gnutella-style flooding). Discuss the trade-offs each makes between lookup efficiency, churn tolerance, and implementation complexity

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2 Part B — Case Study Analysis (30 marks)

Choose one primary case study focus (GFS, HDFS, or Amazon S3) but your discussion must compare all three where relevant.

2.1 1. System Architecture (10 marks)

For each of GFS, HDFS, and Amazon S3, describe:

- Overall architecture (master/worker nodes, metadata management, replication strategy)
- How the system handles fault tolerance and data consistency
- How read/write operations are performed

2.2 2. Cloud Storage Types (10 marks)

Explain the differences between object storage, block storage, and file storage. Classify GFS, HDFS, and Amazon S3 according to these storage models, and justify your classification with technical reasoning.

2.3 3. Comparative Analysis (10 marks)

Create a comparison table evaluating GFS, HDFS, and Amazon S3 across the following dimensions:

Dimension	What to Cover
Scalability	How each system scales with data volume and node count
Consistency Model	Strong vs. eventual consistency guarantees
Fault Tolerance	Replication strategy and recovery from node failure
Typical Use Cases	Real-world workloads each system is optimized for
Cost & Access Model	On-premise vs. cloud-managed, pricing implications

3 Part C — Design & Practical Task (30 marks)

Choose one of the following options.

3.1 Option 1: Distributed File System Design (Theoretical)

Design a simplified distributed file system for a hypothetical company that needs to store and process large log files (each 1–10 GB) generated continuously from thousands of servers.

Your design document (1,000–1,500 words + diagrams) must address:

- Chunking/block size strategy and justification
- Metadata management (centralized master vs. distributed metadata)
- Replication factor and placement strategy
- Consistency model chosen and trade-offs accepted (CAP theorem discussion required)
- Fault detection and recovery mechanism
- A comparison of at least three design trade-offs you made (e.g., consistency vs. availability, throughput vs. latency, simplicity vs. fault tolerance)

3.2 Option 2: Cloud Programming Model Implementation (Practical)

Implement a small working demonstration (in any language: Python/Java/Node.js) of one of the following:

- A MapReduce-style word count or log analysis job (can run locally or simulate distributed workers using threads/processes)
- A simple client that uploads, downloads, and lists objects in a cloud object storage bucket (e.g., using AWS S3 SDK, MinIO, or a local emulator)

Submit your code along with a short report (500–800 words) explaining:

- Which cloud programming model your implementation follows (e.g., MapReduce, serverless/FaaS, actor model)
- How your solution would scale in a real distributed/cloud deployment
- Limitations of your local/simulated version compared to a production cloud environment

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4 Submission Guidelines

- Submit one consolidated PDF report containing Parts A, B, and C.
- If Option 2 is chosen for Part C, include a link to a code repository or a zipped source folder.
- Include a reference list (IEEE or Harvard style) for all external sources cited.
- Diagrams may be hand-drawn (photographed/scanned) or created digitally; label all diagrams clearly.
- Late submissions are subject to the course's standard late-penalty policy.

5 Marking Rubric (Summary)

Component	Criteria	Marks
Part A	Accuracy, depth of explanation, use of examples/diagrams	20
Part B	Technical accuracy, quality of comparison, critical reasoning	30
Part C	Design justification / working implementation, clarity of trade-off analysis	30
Part D	Presentation and Viva	20
Total		100

Deductions: Poor referencing (-5), missing diagrams where required (-5), exceeding word limit by >20% without justification (-5).

6. Deliverables (**VERY IMPORTANT**)

a. INDIVIDUAL WRITTEN REPORT (MANDATORY)

- Each student submits a separate report
- File format: PDF
- Student name & registration number must be clearly mentioned
- Plagiarism between group members will be penalized

! Sharing one report among group members is STRICTLY PROHIBITED

b. **GROUP PRESENTATION (ONE PER GROUP)**

- Maximum 10 slides, prepared collectively, Presented during the viva

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c. **PHYSICAL VIVA (MANDATORY)**

- All students must be physically present
- Viva marks are awarded individually

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8. Physical Viva Session Structure

⏱ Total Duration per Group: 20 Minutes

Activity	Duration
Group Presentation	15 Minutes
Individual Q & A	5 Minutes

Viva Notes

- Questions can be directed at any individual
- Answers must reflect the student's own report

B. Group Presentation – 20 Marks

Criteria	Marks
Technical clarity	10
Visual quality & flow	5

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INTE 22253 Assignment Group Registration-2026 – Fill out form



********* *The deadline for assignment submission is 30th September 2026*********